

3

Continuum Mass and Force Concepts

Here we introduce the notion of a continuum body and discuss how to describe its mass properties, and the various types of forces that may act on it. As we will see, the discussion of internal forces in a continuum body will lead to the notion of a stress tensor field – our first example of a second-order tensor field arising in a physical context. We introduce the basic conditions necessary for the mechanical equilibrium of a continuum body and then derive a corresponding statement in terms of differential equations.

The important ideas in this chapter are: (i) the notion of a mass density field, which enables us to define the mass of an arbitrary part of a body; (ii) the notion of body and surface force fields, which enable us to define the resultant force and torque on an arbitrary part of a body; (iii) the notion of a Cauchy stress field and its relation to surface force fields; (iv) the equations of equilibrium for a body.

3.1 Continuum Bodies

The most basic assumption we make in our study of any material body, whether it be a solid, liquid or gas, is that the material involved can be modeled as a **continuum**: we ignore the atomic nature of the material and assume it is infinitely divisible. This assumption can lead to very effective material models at length scales much longer than typical interatomic spacings. However, at length scales comparable to or smaller than this, it is no longer expected to be valid.

The continuum assumption allows us, at any fixed instant of time, to identify a material body with an open subset B of Euclidean space $\mathbf{E}^3$. In particular, each material particle is identified with a point $\mathbf{x} \in B$. We call B a **placement** or **configuration** of the body in $\mathbf{E}^3$. Unless

mentioned otherwise, we will always assume that B is a regular region of $\mathbf{E}^3$ in the sense defined in Section 2.3. We choose to identify material bodies with open subsets of space purely for mathematical convenience. It would also be reasonable to identify bodies with the closure of such sets.

Physical experience tells us that a body will tend to move and generally change shape under the action of external influences. Thus a given material body will generally assume various configurations as time progresses. For the purposes of this chapter, however, we focus attention on a fixed time and identify a material body with a single configuration B .

3.2 Mass

Mass is a physical property of matter which quantifies its resistance to acceleration. Here we introduce the idea of a mass density field for a continuum body along with the idea of a center of mass. These concepts will appear throughout our developments.

3.2.1 Mass Density

In accordance with the continuum assumption we assume that the mass of a body B is distributed continuously throughout its volume. Moreover, we assume that any subset of B with positive volume has positive mass, and that this mass tends to zero as the volume tends to zero.

To make the above ideas precise, let $\text{mass}[\Omega]$ denote the mass of an arbitrary open subset Ω of B . Then we assume there exists a **mass density field** per unit volume $\rho : B \rightarrow \mathbb{R}$ such that

$$\text{mass}[\Omega] = \int_{\Omega} \rho(\mathbf{x}) \, dV_{\mathbf{x}}.$$

Moreover, we assume $\rho(\mathbf{x}) > 0$ for all $\mathbf{x} \in B$. Here we use $dV_{\mathbf{x}}$ to denote an infinitesimal volume element at $\mathbf{x} \in \Omega$. In particular, the volume of Ω is

$$\text{vol}[\Omega] = \int_{\Omega} dV_{\mathbf{x}}.$$

(In later chapters we will integrate over both $\mathbf{x}$ and $\mathbf{X}$; hence we introduce the good practice of explicitly denoting the variable of integration in surface and volume integrals.)

The mass density field ρ can be formally defined as follows. Let $\mathbf{x}$ be

an arbitrary point in B and let $\Omega_\delta(\mathbf{x})$ denote a family of volumes with the properties that $\text{vol}[\Omega_\delta(\mathbf{x})] \rightarrow 0$ as $\delta \rightarrow 0$ and $\mathbf{x} \in \Omega_\delta(\mathbf{x})$ for every $\delta > 0$. Then

$$\rho(\mathbf{x}) = \lim_{\delta \rightarrow 0} \frac{\text{mass}[\Omega_\delta(\mathbf{x})]}{\text{vol}[\Omega_\delta(\mathbf{x})]}.$$

Our basic assumption on the mass distribution in a continuum is that this limit exists and is positive for each $\mathbf{x}$ in B . Furthermore, we assume this limit is the same for any family $\Omega_\delta(\mathbf{x})$ with the properties described.

3.2.2 Center of Mass

By the **center of mass** of an open subset Ω of B we mean the point

$$\mathbf{x}_{\text{com}}[\Omega] = \frac{1}{\text{mass}[\Omega]} \int_{\Omega} \mathbf{x} \rho(\mathbf{x}) dV_{\mathbf{x}}.$$

This is related to, but in general different from, the **center of volume** or centroid of Ω , which is defined as

$$\mathbf{x}_{\text{cov}}[\Omega] = \frac{1}{\text{vol}[\Omega]} \int_{\Omega} \mathbf{x} dV_{\mathbf{x}}.$$

Notice that $\mathbf{x}_{\text{com}}[\Omega]$ and $\mathbf{x}_{\text{cov}}[\Omega]$ need not be points in B (see Exercises 1 and 2). As discussed below, these points play a special role in determining the resultant torque on a body due to body forces that are uniform in an appropriate sense. The center of mass also plays a special role in the laws of inertia for bodies as discussed later in Chapter 5.

3.3 Force

Mechanical interactions between parts of a body or between a body and its environment are described by forces. Here we describe two basic types of forces: body forces, which are exerted at the interior points of a body, and surface forces, which are exerted on internal surfaces between separate parts of a body, or on external surfaces between the body and its environment.

3.3.1 Body Forces

We use the term **body force** to denote any force that does not arise due to physical contact between bodies. Such a force is the result of action at a distance. A prototypical example is a gravitational force. The body

force, per unit volume, exerted by an external influence on a body B is assumed to be given by a function $\widehat{\mathbf{b}} : B \rightarrow \mathcal{V}$, which we call a **body force field** on B .

Let Ω be an arbitrary open subset of B . Then the resultant force on Ω due to a body force field per unit volume is defined as

$$\mathbf{r}_b[\Omega] = \int_{\Omega} \widehat{\mathbf{b}}(\mathbf{x}) \, dV_{\mathbf{x}},$$

and the resultant torque on Ω , about a point $\mathbf{z}$, is defined as

$$\boldsymbol{\tau}_b[\Omega] = \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \widehat{\mathbf{b}}(\mathbf{x}) \, dV_{\mathbf{x}}.$$

Sometimes it is convenient to introduce a body force field per unit mass defined by

$$\mathbf{b}(\mathbf{x}) = \rho(\mathbf{x})^{-1} \widehat{\mathbf{b}}(\mathbf{x}).$$

In this case, the resultant force becomes

$$\mathbf{r}_b[\Omega] = \int_{\Omega} \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}},$$

and the resultant torque, about a point $\mathbf{z}$, becomes

$$\boldsymbol{\tau}_b[\Omega] = \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \, dV_{\mathbf{x}}.$$

The resultant torque $\boldsymbol{\tau}_b[\Omega]$ can be made to vanish through an appropriate choice of the reference point $\mathbf{z}$. For example, when the body force per unit mass $\mathbf{b}$ is uniform (constant), the resultant torque about the center of mass $\mathbf{x}_{\text{com}}[\Omega]$ vanishes. Similarly, when the body force per unit volume $\widehat{\mathbf{b}}$ is uniform, the resultant torque about the center of volume $\mathbf{x}_{\text{cov}}[\Omega]$ vanishes (see Exercises 4 and 5).

3.3.2 Surface Forces

We use the term **surface force** to denote any force that arises due to physical contact between bodies. When referring to a surface force along an imaginary surface within the interior of a body, we say that the force is **internal**. In contrast, when referring to a surface force along the bounding surface of a body, we say that the force is **external**. Roughly speaking, internal surface forces are what hold a loaded body together: they resist the tendency of one part of a body from being pulled away from another part. External surface forces are contact forces applied to a body by its environment.

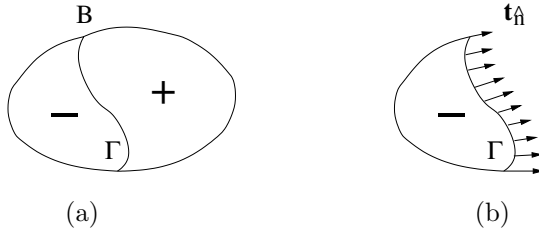


Fig. 3.1 Graphical illustration of internal surface forces. (a) a body B separated into positive and negative parts by an oriented surface Γ (unit normal field $\hat{\mathbf{n}}$ not shown). (b) the action of the positive material upon the negative material is described by a traction field $\mathbf{t}_{\hat{\mathbf{n}}}$ on Γ .

3.3.2.1 Concept of a Traction Field

Let Γ be an arbitrary, oriented surface in B with unit normal field $\hat{\mathbf{n}} : \Gamma \rightarrow \mathcal{V}$. At each point $\mathbf{x}$ the unit normal $\hat{\mathbf{n}}(\mathbf{x})$ defines a positive side and a negative side of Γ as shown in Figure 3.1. The force, per unit area, exerted by material on the positive side upon material on the negative side is assumed to be given by a function $\mathbf{t}_{\hat{\mathbf{n}}} : \Gamma \rightarrow \mathcal{V}$, which we call the **traction** or **surface force field** for Γ . When Γ is part of the bounding surface of B we always choose $\hat{\mathbf{n}}$ to be the outward unit normal field. In this case, the traction field $\mathbf{t}_{\hat{\mathbf{n}}}$ represents the force, per unit area, applied to the surface of B by external means.

The resultant force due to a traction field on an oriented surface Γ is defined as

$$\mathbf{r}_s[\Gamma] = \int_{\Gamma} \mathbf{t}_{\hat{\mathbf{n}}}(\mathbf{x}) dA_{\mathbf{x}},$$

where $dA_{\mathbf{x}}$ represents an infinitesimal surface area element at $\mathbf{x} \in \Gamma$. Similarly, the resultant torque, about a point $\mathbf{z}$, due to a traction field on Γ is defined as

$$\boldsymbol{\tau}_s[\Gamma] = \int_{\Gamma} (\mathbf{x} - \mathbf{z}) \times \mathbf{t}_{\hat{\mathbf{n}}}(\mathbf{x}) dA_{\mathbf{x}}.$$

For examples and applications of these ideas see Exercises 6 and 7.

3.3.2.2 Cauchy's Postulate, Law of Action and Reaction

The theory of surface forces in classical continuum mechanics is based on the following assumption typically referred to as **Cauchy's Postulate**.

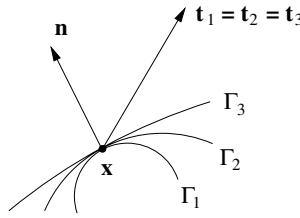


Fig. 3.2 Graphical illustration of Cauchy's postulate: surfaces through a point $\mathbf{x}$ with normal $\mathbf{n}$ at $\mathbf{x}$ share the same traction vector at $\mathbf{x}$.

Axiom 3.1 Dependence of Traction on Surface Geometry. *The traction field $\mathbf{t}_{\hat{\mathbf{n}}}$ on a surface Γ in B depends only pointwise on the unit normal field $\hat{\mathbf{n}}$. In particular, there is a function $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$, where $\mathcal{N} \subset \mathcal{V}$ denotes the set of all unit vectors, such that*

$$\mathbf{t}_{\hat{\mathbf{n}}}(\mathbf{x}) = \mathbf{t}(\hat{\mathbf{n}}(\mathbf{x}), \mathbf{x}).$$

The function $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$ is called the **traction function** for B . $\square$

Thus the traction vector $\mathbf{t}_{\hat{\mathbf{n}}}$ at a point $\mathbf{x}$ depends only on the values of $\mathbf{x}$ and $\hat{\mathbf{n}}(\mathbf{x})$. In particular, it does not depend on the value of $\nabla \hat{\mathbf{n}}(\mathbf{x})$, which describes the curvature of the surface. To see a consequence of this, consider a collection of surfaces Γ_1 , Γ_2 and Γ_3 in B . Suppose that these surfaces all pass through a point $\mathbf{x}$, and that they share a common unit normal $\mathbf{n}$ at $\mathbf{x}$ as shown in Figure 3.2. In particular, $\hat{\mathbf{n}}_i(\mathbf{x}) = \mathbf{n}$ where $\hat{\mathbf{n}}_i$ is the unit normal field for the surface Γ_i . If we let $\mathbf{t}_{\hat{\mathbf{n}}_i}$ be the traction field for surface Γ_i , then

$$\mathbf{t}_{\hat{\mathbf{n}}_1}(\mathbf{x}) = \mathbf{t}_{\hat{\mathbf{n}}_2}(\mathbf{x}) = \mathbf{t}_{\hat{\mathbf{n}}_3}(\mathbf{x}),$$

as depicted in Figure 3.2. Since $\hat{\mathbf{n}}_i(\mathbf{x}) = \mathbf{n}$ the common value of these three traction vectors is $\mathbf{t}(\mathbf{n}, \mathbf{x})$.

The next result shows that the traction function satisfies a certain law of action and reaction. In the following, Ω denotes a regular, open subset of B with boundary $\partial\Omega$ and outward unit normal field $\hat{\mathbf{n}}$.

Result 3.2 Law of Action and Reaction. *Let $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$ be the traction function for a body B . Suppose that $\mathbf{t}(\mathbf{n}, \mathbf{x})$ is continuous and that for any sequence of subsets Ω whose volumes tend to zero*

$$\frac{1}{\text{area}(\partial\Omega)} \int_{\partial\Omega} \mathbf{t}(\hat{\mathbf{n}}(\mathbf{x}), \mathbf{x}) dA_{\mathbf{x}} \rightarrow \mathbf{0} \quad \text{as} \quad \text{vol}(\Omega) \rightarrow 0. \quad (3.1)$$

Then

$$\mathbf{t}(-\mathbf{n}, \mathbf{x}) = -\mathbf{t}(\mathbf{n}, \mathbf{x}), \quad \forall \mathbf{n} \in \mathcal{N}, \mathbf{x} \in B.$$

□

Proof Let $\mathbf{x} \in B$ and $\mathbf{n} \in \mathcal{N}$ be arbitrary, and let $D \subset B$ be a disc of arbitrary, fixed radius centered at $\mathbf{x}$ with normal $\mathbf{n}$. For $\delta > 0$ let $\Omega_\delta \subset B$ be the cylinder with center $\mathbf{x}$, height δ , axis $\mathbf{n}$, end faces $\Gamma_\pm$ parallel to D , and lateral surface Γ_δ as depicted in Figure 3.3(a). Notice that $\text{area}(\Gamma_\delta) \rightarrow 0$ and $\Gamma_\pm \rightarrow D$ as $\delta \rightarrow 0$. Let $\hat{\mathbf{n}}$ be the outward unit normal field on $\partial\Omega_\delta$ and notice that $\hat{\mathbf{n}}(\mathbf{y}) = \pm\mathbf{n}$ (constant) on $\Gamma_\pm$. Considering only the second factor in (3.1), we have

$$\lim_{\delta \rightarrow 0} \int_{\partial\Omega_\delta} \mathbf{t}(\hat{\mathbf{n}}(\mathbf{y}), \mathbf{y}) \, dA_{\mathbf{y}} = \mathbf{0},$$

and since $\partial\Omega_\delta = \Gamma_\delta \cup \Gamma_+ \cup \Gamma_-$, we have

$$\lim_{\delta \rightarrow 0} \left[\int_{\Gamma_\delta} \mathbf{t}(\hat{\mathbf{n}}(\mathbf{y}), \mathbf{y}) \, dA_{\mathbf{y}} + \int_{\Gamma_+} \mathbf{t}(\mathbf{n}, \mathbf{y}) \, dA_{\mathbf{y}} + \int_{\Gamma_-} \mathbf{t}(-\mathbf{n}, \mathbf{y}) \, dA_{\mathbf{y}} \right] = \mathbf{0}.$$

The first term vanishes by continuity (hence boundedness) of $\mathbf{t}(\hat{\mathbf{n}}(\mathbf{y}), \mathbf{y})$ and the fact that $\text{area}(\Gamma_\delta) \rightarrow 0$. Using the fact that $\Gamma_\pm \rightarrow D$ the above limit implies

$$\int_D [\mathbf{t}(\mathbf{n}, \mathbf{y}) + \mathbf{t}(-\mathbf{n}, \mathbf{y})] \, dA_{\mathbf{y}} = \mathbf{0}.$$

Since the radius of D is arbitrary we deduce, by a slight generalization of the Localization Theorem (Result 2.19), that the integrand must vanish at the center of D . Thus

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) + \mathbf{t}(-\mathbf{n}, \mathbf{x}) = \mathbf{0},$$

which establishes the result. □

Remarks:

- (1) The above result implies that the traction exerted by material on the positive side upon material on the negative side of an oriented surface is equal and opposite, at each point, to the traction exerted by material on the negative side upon material on the positive side.

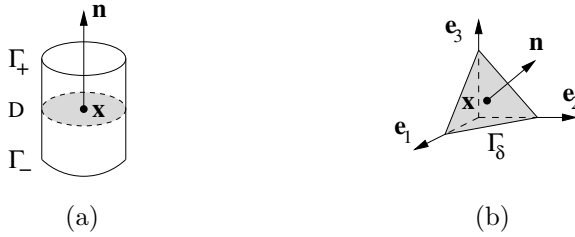


Fig. 3.3 Illustration of the regions used in the proofs of Results 3.2 and 3.3. (a) cylinder defined by a disc D . (b) tetrahedron defined by a triangular region Γ_δ .

- (2) The condition (3.1) states that the resultant surface force on a body, divided by the surface area, tends to zero as the body volume tends to zero. This condition is consistent with the axiom on balance of linear momentum (Axiom 5.2) under the assumption that the body force and acceleration fields are bounded. Throughout our developments we always assume this to be the case.
- (3) The proof of Result 3.2 does not make complete use of (3.1). In particular, the proof makes use of only the second factor in (3.1), which implies that the resultant surface force tends to zero as the body volume tends to zero. Complete use of (3.1) will be made in Result 3.3 below. $\square$

3.3.3 The Stress Tensor

Using Result 3.2 we can determine the specific way in which the traction function $\mathbf{t}(\mathbf{n}, \mathbf{x})$ depends on $\mathbf{n}$. The following result is typically referred to as **Cauchy's Theorem** and will be fundamental throughout our developments.

Result 3.3 Existence of Stress Tensor. *Let $\mathbf{t} : \mathcal{N} \times B \rightarrow \mathcal{V}$ be the traction function for a body B and suppose it satisfies the conditions of Result 3.2. Then $\mathbf{t}(\mathbf{n}, \mathbf{x})$ is linear in $\mathbf{n}$, that is, for each $\mathbf{x} \in B$ there is a second-order tensor $\mathbf{S}(\mathbf{x}) \in \mathcal{V}^2$ such that*

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = \mathbf{S}(\mathbf{x})\mathbf{n}.$$

The field $\mathbf{S} : B \rightarrow \mathcal{V}^2$ is called the **Cauchy stress field** for B . $\square$

Proof Let $\{\mathbf{e}_j\}$ be an arbitrary frame, $\mathbf{x} \in B$ an arbitrary point and consider any $\mathbf{n} \in \mathcal{N}$ such that $n_j = \mathbf{n} \cdot \mathbf{e}_j > 0$ for each j . For $\delta > 0$ let $\Gamma_\delta \subset B$ denote a triangular region with center $\mathbf{x}$, normal $\mathbf{n}$, maximum edge length δ , and each edge in a different coordinate plane as depicted in Figure 3.3(b). Moreover, let $\Omega_\delta \subset B$ be the tetrahedron bounded by Γ_δ and the three coordinate planes. In particular, Ω_δ has three faces Γ_j with outward normals $-\mathbf{e}_j$, one face Γ_δ with outward normal $\mathbf{n}$, and $\text{vol}(\Omega_\delta) \rightarrow 0$ as $\delta \rightarrow 0$. Let $\hat{\mathbf{n}}$ be the outward unit normal field on $\partial\Omega_\delta$ and notice that $\hat{\mathbf{n}}(\mathbf{y}) = -\mathbf{e}_j$ (constant) on Γ_j and $\hat{\mathbf{n}}(\mathbf{y}) = \mathbf{n}$ (constant) on Γ_δ . By (3.1), we have

$$\lim_{\delta \rightarrow 0} \frac{1}{\text{area}(\partial\Omega_\delta)} \int_{\partial\Omega_\delta} \mathbf{t}(\hat{\mathbf{n}}(\mathbf{y}), \mathbf{y}) dA_{\mathbf{y}} = \mathbf{0},$$

and since $\partial\Omega_\delta = \Gamma_\delta \cup \Gamma_1 \cup \Gamma_2 \cup \Gamma_3$, we have

$$\lim_{\delta \rightarrow 0} \frac{1}{\text{area}(\partial\Omega_\delta)} \left[\int_{\Gamma_\delta} \mathbf{t}(\mathbf{n}, \mathbf{y}) dA_{\mathbf{y}} + \sum_{j=1}^3 \int_{\Gamma_j} \mathbf{t}(-\mathbf{e}_j, \mathbf{y}) dA_{\mathbf{y}} \right] = \mathbf{0}. \quad (3.2)$$

Since each face Γ_j can be linearly mapped onto Γ_δ with constant Jacobian $n_j > 0$, we have $\text{area}(\Gamma_j) = n_j \text{area}(\Gamma_\delta)$ and

$$\text{area}(\partial\Omega_\delta) = \text{area}(\Gamma_\delta) + \sum_{j=1}^3 \text{area}(\Gamma_j) = \lambda \text{area}(\Gamma_\delta),$$

where $\lambda = 1 + \sum_{j=1}^3 n_j$ is independent of δ . Substituting this result into (3.2), multiplying by λ , and changing variables between Γ_j and Γ_δ , we obtain

$$\lim_{\delta \rightarrow 0} \frac{1}{\text{area}(\Gamma_\delta)} \int_{\Gamma_\delta} \left[\mathbf{t}(\mathbf{n}, \mathbf{y}) + \sum_{j=1}^3 \mathbf{t}(-\mathbf{e}_j, \mathbf{y}) n_j \right] dA_{\mathbf{y}} = \mathbf{0}.$$

By the Mean Value Theorem for integrals, since the integrand is continuous and Γ_δ is shrinking to the point $\mathbf{x}$, the limit on the left-hand side is the integrand at $\mathbf{x}$. Thus we have

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) + \sum_{j=1}^3 \mathbf{t}(-\mathbf{e}_j, \mathbf{x}) n_j = \mathbf{0}.$$

Using Result 3.2, and employing the summation convention, this can be written as

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = \mathbf{t}(\mathbf{e}_j, \mathbf{x}) n_j = \left(\mathbf{t}(\mathbf{e}_j, \mathbf{x}) \otimes \mathbf{e}_j \right) \mathbf{n} = \mathbf{S}(\mathbf{x}) \mathbf{n}, \quad (3.3)$$

where

$$\mathbf{S}(\mathbf{x}) = \mathbf{t}(\mathbf{e}_j, \mathbf{x}) \otimes \mathbf{e}_j. \quad (3.4)$$

Thus for arbitrary $\mathbf{x}$ there exists a second-order tensor $\mathbf{S}(\mathbf{x})$ such that $\mathbf{t}(\mathbf{n}, \mathbf{x}) = \mathbf{S}(\mathbf{x})\mathbf{n}$ for all $\mathbf{n}$ such that $\mathbf{n} \cdot \mathbf{e}_j > 0$.

The above result can be extended to other unit vectors. For example, if $\mathbf{n}$ satisfies $\mathbf{n} \cdot \mathbf{e}_1 < 0$, with the other components being positive, then we can introduce a new frame $\{\mathbf{e}'_j\}$ such that $\mathbf{e}'_1 = -\mathbf{e}_1$, $\mathbf{e}'_2 = \mathbf{e}_3$ and $\mathbf{e}'_3 = \mathbf{e}_2$. In this frame $\mathbf{n}$ satisfies $\mathbf{n} \cdot \mathbf{e}'_j > 0$ and the above arguments apply giving

$$\mathbf{t}(\mathbf{n}, \mathbf{x}) = \left(\mathbf{t}(\mathbf{e}'_j, \mathbf{x}) \otimes \mathbf{e}'_j \right) \mathbf{n}.$$

However, by definition of $\{\mathbf{e}'_j\}$ and Result 3.2, we have, still employing the summation convention

$$\mathbf{t}(\mathbf{e}'_j, \mathbf{x}) \otimes \mathbf{e}'_j = \mathbf{t}(\mathbf{e}_j, \mathbf{x}) \otimes \mathbf{e}_j.$$

Thus (3.3) also holds for all $\mathbf{n}$ with $\mathbf{n} \cdot \mathbf{e}_1 < 0$ and $\mathbf{n} \cdot \mathbf{e}_j > 0$ ($j = 2, 3$). Continuing in this way we find that $\mathbf{t}(\mathbf{n}, \mathbf{x}) = \mathbf{S}(\mathbf{x})\mathbf{n}$ for all $\mathbf{n}$ such that $\mathbf{n} \cdot \mathbf{e}_j \neq 0$ ($j = 1, 2, 3$). By the assumed continuity of $\mathbf{t}(\mathbf{n}, \mathbf{x})$ this result then extends to all $\mathbf{n}$. $\square$

Remarks:

- (1) Throughout the remainder of our developments we will denote normal fields by $\mathbf{n}(\mathbf{x})$ rather than $\hat{\mathbf{n}}(\mathbf{x})$. Thus, by Axiom 3.1 and Result 3.3, the traction field on a surface with normal $\mathbf{n}(\mathbf{x})$ is

$$\mathbf{t}(\mathbf{n}(\mathbf{x}), \mathbf{x}) = \mathbf{S}(\mathbf{x})\mathbf{n}(\mathbf{x}).$$

When there is no cause for confusion, we will abbreviate this relation by $\mathbf{t}(\mathbf{x}) = \mathbf{S}(\mathbf{x})\mathbf{n}$, or more simply

$$\mathbf{t} = \mathbf{S}\mathbf{n}.$$

In components in any frame $\{\mathbf{e}_i\}$ this reads $t_i = S_{ij}n_j$.

- (2) The nine components of the stress tensor $\mathbf{S}(\mathbf{x})$ can be understood as the components of the three traction vectors $\mathbf{t}(\mathbf{e}_j, \mathbf{x})$ on the coordinate planes at $\mathbf{x}$. In particular, substituting $\mathbf{t}(\mathbf{e}_j, \mathbf{x}) = t_i(\mathbf{e}_j, \mathbf{x})\mathbf{e}_i$ into (3.4) we get

$$\mathbf{S}(\mathbf{x}) = S_{ij}(\mathbf{x})\mathbf{e}_i \otimes \mathbf{e}_j \quad \text{where} \quad S_{ij}(\mathbf{x}) = t_i(\mathbf{e}_j, \mathbf{x}).$$

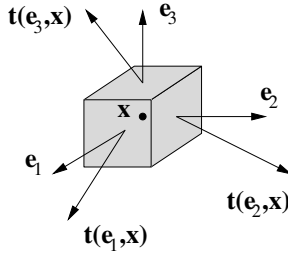


Fig. 3.4 Illustration of the traction vectors on the coordinate planes with normals $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ at $\mathbf{x}$. These traction vectors can be interpreted as the surface forces (per unit area) on an infinitesimal cube centered at $\mathbf{x}$.

Thus the traction vectors on the coordinate planes with normals $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$ at $\mathbf{x}$ are (see Figure 3.4)

$$\mathbf{t}(\mathbf{e}_1, \mathbf{x}) = t_i(\mathbf{e}_1, \mathbf{x})\mathbf{e}_i = S_{i1}(\mathbf{x})\mathbf{e}_i,$$

$$\mathbf{t}(\mathbf{e}_2, \mathbf{x}) = t_i(\mathbf{e}_2, \mathbf{x})\mathbf{e}_i = S_{i2}(\mathbf{x})\mathbf{e}_i,$$

$$\mathbf{t}(\mathbf{e}_3, \mathbf{x}) = t_i(\mathbf{e}_3, \mathbf{x})\mathbf{e}_i = S_{i3}(\mathbf{x})\mathbf{e}_i.$$

□

3.4 Equilibrium

Here we state an axiom which provides necessary conditions for a continuum body to be in mechanical equilibrium; that is, at rest in a fixed frame for Euclidean space $\mathbf{E}^3$. We then derive a local form of the necessary conditions in terms of differential equations.

3.4.1 Preliminaries

Consider a continuum body in Euclidean space $\mathbf{E}^3$. Suppose that at a given instant of time it is motionless and has a configuration B_0 as shown in Figure 3.5(a). Suppose next that the body is subject to an external traction field and a body force field such that the body changes shape and comes to rest in a configuration B as shown in Figure 3.5(b). In the configuration B we denote the mass density field per unit volume by $\rho : B \rightarrow \mathbf{R}$, the applied external traction field per unit area by $\mathbf{h} : \partial B \rightarrow \mathcal{V}$ and the body force field per unit mass by $\mathbf{b} : B \rightarrow \mathcal{V}$. We implicitly assume that ρ , $\mathbf{h}$ and $\mathbf{b}$ do not depend on time.

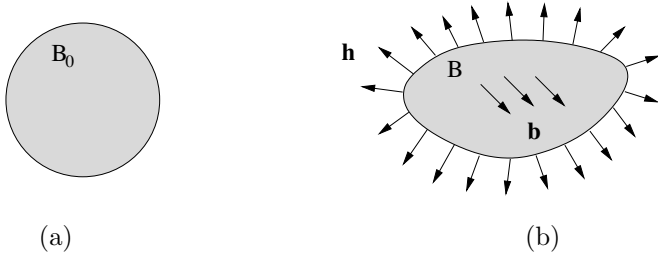


Fig. 3.5 Two equilibrium configurations of a material body. (a) equilibrium in the absence of external loads. (b) equilibrium in the presence of a traction $\mathbf{h}$ and a body force $\mathbf{b}$.

3.4.2 Necessary Conditions

Let Ω be any open subset of B and let $\mathbf{t} : \partial\Omega \rightarrow \mathcal{V}$ be the traction field acting on its bounding surface, with orientation determined by the unit outward normal field. (Notice that $\mathbf{t} = \mathbf{h}$ in the special case when $\Omega = B$.) The resultant force on Ω due to body and surface forces is

$$\mathbf{r}[\Omega] = \mathbf{r}_b[\Omega] + \mathbf{r}_s[\partial\Omega] = \int_{\Omega} \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{t}(\mathbf{x}) dA_{\mathbf{x}}, \quad (3.5)$$

and the resultant torque on Ω , about a point $\mathbf{z}$, due to body and surface forces is

$$\begin{aligned} \boldsymbol{\tau}[\Omega] &= \boldsymbol{\tau}_b[\Omega] + \boldsymbol{\tau}_s[\partial\Omega] \\ &= \int_{\Omega} (\mathbf{x} - \mathbf{z}) \times \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) dV_{\mathbf{x}} + \int_{\partial\Omega} (\mathbf{x} - \mathbf{z}) \times \mathbf{t}(\mathbf{x}) dA_{\mathbf{x}}. \end{aligned} \quad (3.6)$$

Necessary conditions for equilibrium, which actually follow from more general axioms discussed later (see Axiom 5.2), can be stated as follows.

Axiom 3.4 Conditions for Equilibrium. *If a body in a configuration B is in mechanical equilibrium, then the resultant force and resultant torque about any fixed point, say the origin, must vanish for every open subset Ω of B . That is*

$$\left. \begin{aligned} \mathbf{r}[\Omega] &= \int_{\Omega} \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{t}(\mathbf{x}) dA_{\mathbf{x}} = \mathbf{0} \\ \boldsymbol{\tau}[\Omega] &= \int_{\Omega} \mathbf{x} \times \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) dV_{\mathbf{x}} + \int_{\partial\Omega} \mathbf{x} \times \mathbf{t}(\mathbf{x}) dA_{\mathbf{x}} = \mathbf{0} \end{aligned} \right\} \quad \forall \Omega \subseteq B. \quad (3.7)$$

□

The freedom in the choice of reference point for the resultant torque is a consequence of (3.7)₁. In particular, if $\mathbf{r}[\Omega]$ vanishes, then $\boldsymbol{\tau}[\Omega]$ as defined in (3.6) is independent of $\mathbf{z}$ (see Exercise 8).

3.4.3 Local Equations

Here we derive a set of local or pointwise equations corresponding to the necessary conditions in Axiom 3.4. In view of Result 3.3, these equations naturally involve the Cauchy stress field $\mathbf{S}$.

Result 3.5 Local Equilibrium Equations. *If the Cauchy stress field $\mathbf{S}$ is continuously differentiable, and the density field ρ and body force field $\mathbf{b}$ are continuous, then the equilibrium conditions in (3.7) are equivalent to*

$$\left. \begin{aligned} (\nabla \cdot \mathbf{S})(\mathbf{x}) + \rho(\mathbf{x})\mathbf{b}(\mathbf{x}) &= \mathbf{0} \\ \mathbf{S}^T(\mathbf{x}) &= \mathbf{S}(\mathbf{x}) \end{aligned} \right\} \quad \forall \mathbf{x} \in B, \quad (3.8)$$

or in components

$$\left. \begin{aligned} S_{ij,j}(\mathbf{x}) + \rho(\mathbf{x})b_i(\mathbf{x}) &= 0 \\ S_{ij}(\mathbf{x}) &= S_{ji}(\mathbf{x}) \end{aligned} \right\} \quad \forall \mathbf{x} \in B.$$

□

Proof To establish (3.8)₁ we use the definition of the Cauchy stress field (Result 3.3) to rewrite (3.7)₁ as

$$\int_{\partial\Omega} \mathbf{S}\mathbf{n} \, dA_{\mathbf{x}} + \int_{\Omega} \rho\mathbf{b} \, dV_{\mathbf{x}} = \mathbf{0},$$

where for convenience we omit the explicit dependence on $\mathbf{x}$ of the integrands, and $\mathbf{n}$ denotes the outward unit normal field on $\partial\Omega$. Using the Tensor Divergence Theorem (Result 2.17) we obtain

$$\int_{\Omega} (\nabla \cdot \mathbf{S} + \rho\mathbf{b}) \, dV_{\mathbf{x}} = \mathbf{0}.$$

Since the above equation must hold for an arbitrary region Ω in B , and the integrand is continuous by assumption, we apply the Localization Theorem (Result 2.19) to conclude the expression in (3.8)₁ (see Exercise 9).

To establish $(3.8)_2$ we use the definition of the Cauchy stress field to rewrite $(3.7)_2$ as

$$\int_{\partial\Omega} \mathbf{x} \times (\mathbf{S}\mathbf{n}) dA_{\mathbf{x}} + \int_{\Omega} \mathbf{x} \times \rho \mathbf{b} dV_{\mathbf{x}} = \mathbf{0},$$

and then use the result $\rho \mathbf{b} = -\nabla \cdot \mathbf{S}$ from $(3.8)_1$ to rewrite the above equation as

$$\int_{\partial\Omega} \mathbf{x} \times (\mathbf{S}\mathbf{n}) dA_{\mathbf{x}} - \int_{\Omega} \mathbf{x} \times (\nabla \cdot \mathbf{S}) dV_{\mathbf{x}} = \mathbf{0}. \quad (3.9)$$

To simplify (3.9) we define a second-order tensor field $\mathbf{R} = R_{il}\mathbf{e}_i \otimes \mathbf{e}_l$ by

$$R_{il} = \epsilon_{ijk} x_j S_{kl}.$$

This tensor field has the property that $\mathbf{R}\mathbf{n} = \mathbf{x} \times (\mathbf{S}\mathbf{n})$, and allows (3.9) to be written as

$$\int_{\partial\Omega} \mathbf{R}\mathbf{n} dA_{\mathbf{x}} - \int_{\Omega} \mathbf{x} \times (\nabla \cdot \mathbf{S}) dV_{\mathbf{x}} = \mathbf{0}. \quad (3.10)$$

An application of the Tensor Divergence Theorem (Result 2.17) on the first term in (3.10) yields

$$\int_{\Omega} \nabla \cdot \mathbf{R} - \mathbf{x} \times (\nabla \cdot \mathbf{S}) dV_{\mathbf{x}} = \mathbf{0}.$$

Since this equation must hold for any subset Ω of B , and the integrand is continuous by assumption, we deduce by the Localization Theorem (Result 2.19) that

$$\nabla \cdot \mathbf{R} - \mathbf{x} \times (\nabla \cdot \mathbf{S}) = \mathbf{0}, \quad \forall \mathbf{x} \in B,$$

which in components becomes

$$(\epsilon_{ijk} x_j S_{kl}),_l - \epsilon_{ijk} x_j S_{kl,l} = 0, \quad \forall \mathbf{x} \in B. \quad (3.11)$$

The left-hand side of this equation may be simplified

$$\begin{aligned} (\epsilon_{ijk} x_j S_{kl}),_l - \epsilon_{ijk} x_j S_{kl,l} &= \epsilon_{ijk} x_{j,l} S_{kl} + \epsilon_{ijk} x_j S_{kl,l} - \epsilon_{ijk} x_j S_{kl,l} \\ &= \epsilon_{ijk} x_{j,l} S_{kl} \\ &= \epsilon_{ijk} \delta_{jl} S_{kl} \\ &= \epsilon_{ijk} S_{kj}. \end{aligned}$$

Thus (3.11) gives $\epsilon_{ijk} S_{kj} = 0$, which implies $\epsilon_{ijk} S_{kj} + \epsilon_{ikj} S_{jk} = 0$, since j and k are dummy indices. By properties of the permutation symbol we can rewrite this last expression as

$$\epsilon_{ijk} [S_{kj} - S_{jk}] = 0.$$

Taking (i, j, k) to be distinct, and allowing $i = 1, 2, 3$, we conclude that $S_{kj} = S_{jk}$ ($1 \leq j, k \leq 3$) for each point $\mathbf{x} \in B$, which is the statement in $(3.8)_2$.

The integral equations (3.7) can similarly be derived from the local equations (3.8). In particular, we need only integrate (3.8) over an arbitrary open subset $\Omega \subset B$ and reverse the order of the arguments above. $\square$

Remarks:

- (1) The symmetry property of the Cauchy stress field holds even when a body is not in equilibrium (see Chapter 5).
- (2) Some texts define a stress tensor $\mathbf{T}$ which is the transpose of the Cauchy stress field $\mathbf{S}$ as defined here; that is, $\mathbf{T} = \mathbf{S}^T$. However, because of the symmetry of the Cauchy stress field this distinction is immaterial in practice.
- (3) The local equations (3.8) do not completely determine the Cauchy stress field for a body in equilibrium. In particular, we have three partial differential equations in $(3.8)_1$ and three independent algebraic equations in $(3.8)_2$ with which to determine the nine unknown components of $\mathbf{S}$. This shortage of equations will arise frequently in later chapters. It is addressed by adding **constitutive equations** which characterize the specific material properties of a body.
- (4) The traction field $\mathbf{h}$ on the bounding surface ∂B represents the surface force, per unit area, exerted by the environment on B . Thus by Result 3.3 we obtain $\mathbf{S}\mathbf{n} = \mathbf{h}$ for all $\mathbf{x} \in \partial B$, where $\mathbf{n}$ is the outward unit normal field on ∂B . This provides **boundary conditions** for the partial differential equations in $(3.8)_1$ (see Exercise 12).
- (5) In deriving the equilibrium equations we have assumed continuity of ρ and $\mathbf{b}$, and continuous differentiability of $\mathbf{S}$. In practice, establishing such regularity properties for the field equations of continuum mechanics is an important part of the subject. Throughout our developments we will assume, without explicit reference, that the fields in question are sufficiently regular to pass from integral laws to differential equations via localization. $\square$

3.5 Basic Stress Concepts

Here we describe various states of stress that may exist at a point in a continuum body and define principal stress values and stress directions. We introduce the concepts of normal and shear stresses at a point on a surface with a prescribed normal vector, and then determine those surfaces that experience maximum normal and shear stresses. We also introduce a decomposition of the stress field into spherical and deviatoric parts and discuss their physical interpretations.

3.5.1 Simple States of Stress

When the stress tensor $\mathbf{S}$ is of a particular form at a point $\mathbf{x}$ in a continuum body, we often say that the stress field is in a particular **state** at the point $\mathbf{x}$. For example, if at a point $\mathbf{x}$ the stress tensor $\mathbf{S}$ is of the form

$$\mathbf{S} = -\pi \mathbf{I},$$

where π is a scalar, we say that a **spherical** or **Eulerian** state of stress exists at $\mathbf{x}$. For this stress state the traction on any surface with normal $\mathbf{n}$ at $\mathbf{x}$ is given by

$$\mathbf{t} = \mathbf{S}\mathbf{n} = -\pi\mathbf{n},$$

which shows that the traction is always normal to the surface. For example, assuming $\mathbf{S}$ is constant in a neighborhood of $\mathbf{x}$, the traction field on the surface of a small sphere centered at $\mathbf{x}$ would be everywhere normal to the sphere, as illustrated in Figure 3.6(a) (see Exercise 13).

The state of stress at a point $\mathbf{x}$ is said to be **uniaxial** if there exists a unit vector $\boldsymbol{\gamma}$ and a scalar σ such that

$$\mathbf{S} = \sigma \boldsymbol{\gamma} \otimes \boldsymbol{\gamma}.$$

Such a state is called a pure tension if $\sigma > 0$, and a pure compression if $\sigma < 0$. For a uniaxial stress state the traction on any surface with normal $\mathbf{n}$ at $\mathbf{x}$ is given by

$$\mathbf{t} = \mathbf{S}\mathbf{n} = (\boldsymbol{\gamma} \cdot \mathbf{n})\sigma\boldsymbol{\gamma}.$$

Notice that the traction is always parallel to $\boldsymbol{\gamma}$ and vanishes when $\mathbf{n}$ is orthogonal to $\boldsymbol{\gamma}$. For example, if we assume $\mathbf{S}$ is constant in a neighborhood of $\mathbf{x}$ and consider a small box centered at $\mathbf{x}$ having one axis aligned with $\boldsymbol{\gamma}$, then the traction field on the surface of the box would

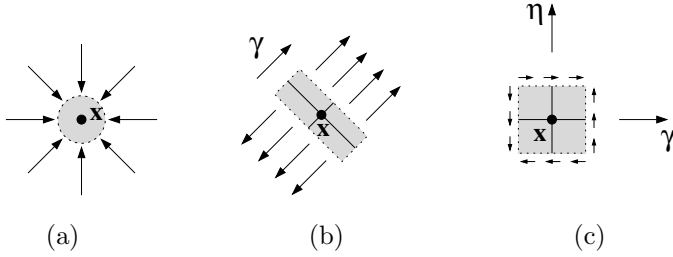


Fig. 3.6 Graphical illustration of three simple states of stress at a point in a continuum body: (a) spherical, (b) uniaxial and (c) pure shear.

be non-zero only on those faces with normal $\mathbf{n} = \pm\gamma$, as illustrated in Figure 3.6(b).

If at a point $\mathbf{x}$ there exists a pair of orthogonal unit vectors γ and η and a scalar τ such that

$$\mathbf{S} = \tau (\gamma \otimes \eta + \eta \otimes \gamma),$$

we say that a state of **pure shear** exists at $\mathbf{x}$. For this stress state the traction on any surface with normal $\mathbf{n}$ at $\mathbf{x}$ is given by

$$\mathbf{t} = \mathbf{S}\mathbf{n} = (\eta \cdot \mathbf{n})\tau\gamma + (\gamma \cdot \mathbf{n})\tau\eta.$$

Notice that $\mathbf{t} = \tau\gamma$ when $\mathbf{n} = \eta$, and $\mathbf{t} = \tau\eta$ when $\mathbf{n} = \gamma$. For example, if we assume $\mathbf{S}$ is constant in a neighborhood of $\mathbf{x}$ and consider a small box centered at $\mathbf{x}$ having one axis aligned with γ and another with η , then the traction field on the surface of the box would be non-zero only on those faces with normal $\mathbf{n} = \pm\gamma$ and $\pm\eta$, as illustrated in Figure 3.6(c).

If at a point $\mathbf{x}$ there exists a pair of orthogonal unit vectors γ and η such that the matrix representation of $\mathbf{S}$ in the basis $\mathbf{e}_1 = \gamma$, $\mathbf{e}_2 = \eta$ and $\mathbf{e}_3 = \gamma \times \eta$ is of the form

$$[\mathbf{S}] = \begin{pmatrix} S_{11} & S_{12} & 0 \\ S_{21} & S_{22} & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

we say that a state of **plane stress** exists at $\mathbf{x}$. It is straightforward to verify that uniaxial and pure shear stress states are special cases of a plane stress state (see Exercise 14).

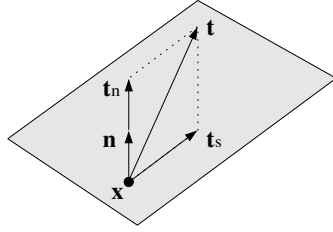


Fig. 3.7 Decomposition of the traction vector into normal and shear tractions at a point $\mathbf{x}$. The shear traction is tangent to the surface and the normal traction is normal to the surface.

3.5.2 Principal, Normal and Shear Stresses

Given a point $\mathbf{x}$ in a continuum body we call the eigenvalues of $\mathbf{S}$ the **principal stresses** and the corresponding eigenvectors the **principal stress directions** at $\mathbf{x}$. Because the stress tensor is symmetric, we note that there exist three mutually perpendicular principal directions and three corresponding principal stresses for each point $\mathbf{x}$ (see Result 1.9).

If we let $\mathbf{e}$ be a principal direction with corresponding principal stress σ , then the traction vector on a surface with normal $\mathbf{e}$ at $\mathbf{x}$ is given by

$$\mathbf{t} = \mathbf{S}\mathbf{e} = \sigma\mathbf{e}.$$

Hence the traction vector is itself in the direction $\mathbf{e}$.

Consider an arbitrary surface with unit normal $\mathbf{n}$ at $\mathbf{x}$. Then the traction vector can be decomposed into the sum of two parts: a **normal traction**

$$\mathbf{t}_n = (\mathbf{t} \cdot \mathbf{n})\mathbf{n},$$

and a **shear traction**

$$\mathbf{t}_s = \mathbf{t} - (\mathbf{t} \cdot \mathbf{n})\mathbf{n}.$$

In particular, we have $\mathbf{t} = \mathbf{t}_n + \mathbf{t}_s$, as illustrated in Figure 3.7. We call $\sigma_n = |\mathbf{t}_n|$ the **normal stress** and $\sigma_s = |\mathbf{t}_s|$ the **shear stress** on the surface with normal $\mathbf{n}$ at $\mathbf{x}$.

If we consider a surface with normal $\mathbf{n} = \mathbf{e}$, where $\mathbf{e}$ is a principal direction with corresponding principal stress σ , then we find that $\mathbf{t}_n = \sigma\mathbf{e}$ and $\mathbf{t}_s = \mathbf{0}$. Hence the normal stress on such a surface is $\sigma_n = |\sigma|$ and the shear stress is zero (see Exercises 15 and 16). In particular, the shear stress on a surface is zero if and only if $\mathbf{n}$ is a principal direction.

3.5.3 Maximum Normal and Shear Stresses

Given a point $\mathbf{x}$ in a continuum body it is of interest to know what families of surfaces through $\mathbf{x}$ experience a maximum normal stress or a maximum shear stress. That is, we would like to know for what normal direction $\mathbf{n}$ at $\mathbf{x}$ is σ_n a maximum; similarly, for what normal direction is σ_s a maximum.

To begin, we denote the principal stresses at the point $\mathbf{x}$ by σ_i and we denote the corresponding principal directions by $\mathbf{e}_i$. For simplicity, we assume that the principal stresses are distinct and ordered in the sense that

$$\sigma_1 > \sigma_2 > \sigma_3. \quad (3.12)$$

When the principal stresses are not distinct as above, there may not be isolated directions $\mathbf{n}$ for which σ_n or σ_s achieve maximum values. For example, when $\sigma_i = \alpha$ ($i = 1, 2, 3$), we can verify that $\sigma_n = |\alpha|$ and $\sigma_s = 0$ for all $\mathbf{n}$. Thus there is no isolated direction $\mathbf{n}$ for which σ_n or σ_s achieves a maximum value. The next result shows that, when the principal stresses are distinct as in (3.12), there are isolated direction pairs $(\mathbf{n}, -\mathbf{n})$ for which σ_n and σ_s achieve maximum values.

Result 3.6 Maximal Normal and Shear Stresses. *Consider a point $\mathbf{x}$ in a continuum body and assume the principal stresses are distinct and ordered as in (3.12). Then:*

- (1) *the maximum value of σ_n is $|\sigma_k|$ and this value is achieved for pairs $\mathbf{n} = \pm \mathbf{e}_k$, where $k = 1$ and/or $k = 3$ depending on which value of k achieves the maximum in the definition*

$$|\sigma_k| = \max\{|\sigma_1|, |\sigma_3|\},$$

- (2) *the maximum value of σ_s is $\frac{1}{2}|\sigma_1 - \sigma_3|$ and this value is achieved for the two pairs*

$$\mathbf{n} = \pm \frac{1}{\sqrt{2}}(\mathbf{e}_1 + \mathbf{e}_3), \quad \pm \frac{1}{\sqrt{2}}(\mathbf{e}_1 - \mathbf{e}_3).$$

□

Proof See Exercises 17, 18.

□

3.5.4 Spherical and Deviatoric Stress Tensors

At any point $\mathbf{x}$ in a continuum body the Cauchy stress tensor can be decomposed into the sum of two parts: a **spherical stress tensor**

$$\mathbf{S}_S = -p\mathbf{I},$$

and a **deviatoric stress tensor**

$$\mathbf{S}_D = \mathbf{S} + p\mathbf{I},$$

where $p = -\frac{1}{3} \operatorname{tr} \mathbf{S}$ is called the **pressure**. In particular, we have $\mathbf{S} = \mathbf{S}_S + \mathbf{S}_D$.

The spherical stress tensor $\mathbf{S}_S$ is completely determined by the pressure p . Noting that

$$-p = \frac{1}{3} \operatorname{tr} \mathbf{S} = \frac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3),$$

we see that $-p$ is the arithmetic average of the three principal stresses at a point $\mathbf{x}$. Equivalently, $-p$ can be interpreted as the average of the normal component of traction $\mathbf{t} \cdot \mathbf{n}$, where the average is taken over all possible directions $\mathbf{n}$ at $\mathbf{x}$ (see Exercise 19). Roughly speaking, the spherical stress is that part of the overall stress that tends to change the volume of a body without changing its shape: a round ball of uniform material subject to a uniform pressure should experience a change of volume while retaining its round shape. Notice that the sign convention for pressure is such that $p > 0$ corresponds to a compression.

The deviatoric stress is that part of the overall stress which is complementary to the spherical stress. In particular, the deviatoric stress tensor contains information about shear stresses, whereas the spherical stress tensor contains information about normal stresses. Roughly speaking, the deviatoric stress is that part of the overall stress that tends to change the shape of a body while producing little or no change in volume: a cube of uniform material subject to a state of pure shear should lose its cubical shape while experiencing only a small change in volume.

By virtue of its definition the deviatoric stress tensor has the property that

$$\operatorname{tr} \mathbf{S}_D = 0. \quad (3.13)$$

Hence the first principal invariant $I_1(\mathbf{S}_D)$ is zero (see Section 1.3.9). The remaining principal invariants $I_2(\mathbf{S}_D)$ and $I_3(\mathbf{S}_D)$ are usually denoted by $-J_2(\mathbf{S})$ and $J_3(\mathbf{S})$. In particular, by (3.13), the symmetry of $\mathbf{S}_D$, and the definition of the principal invariants, we obtain

$$J_2(\mathbf{S}) = -I_2(\mathbf{S}_D) = \frac{1}{2} \mathbf{S}_D : \mathbf{S}_D \quad \text{and} \quad J_3(\mathbf{S}) = I_3(\mathbf{S}_D) = \det \mathbf{S}_D.$$

These invariants play a central role in the formulation of so-called *yield conditions* in theories describing the plastic deformation of isotropic materials at small strains (see Exercise 20).

Bibliographic Notes

Most of the material presented here on bodies, mass and force can be found in standard texts in continuum mechanics, for example Gurtin (1981), Chadwick (1976), Mase (1970) and Malvern (1969). A rigorous, axiomatic treatment of these primitive concepts can be found in Truesdell (1991). In some applications it is useful to introduce the additional primitive concept of a couple, which can be described as a torque that is not the moment of a force. For a discussion of couples in continuum mechanics see Malvern (1969).

The theory of internal surface forces in a continuum body as described here consists of four main parts: (i) the existence of a traction field on an oriented surface, which is referred to as the Euler–Cauchy Cut Principle; (ii) Cauchy’s Postulate, which describes the manner in which the traction field is dependent on the surface geometry; (iii) the law of action and reaction for traction fields, which is typically referred to as Cauchy’s Lemma; (iv) Cauchy’s Theorem, which establishes the existence of the stress tensor. A rigorous account of all these aspects of the theory is contained in Truesdell (1991), and a more concise account is given in Gurtin (1981).

The equilibrium equations presented here are special cases of the balance laws which govern the rate of change of total linear and angular momentum for a continuum body. These laws are typically referred to as Euler’s or Cauchy’s laws of motion depending on whether they are in integral or local form, respectively. In particular, the integral form of the equilibrium equations given in Axiom 3.4 arises when the rates of change of linear and angular momentum in Euler’s laws of motion in Axiom 5.2 are set to zero. A discussion of the momentum balance laws will be given in Chapter 5. For a detailed treatment and historical overview of Euler’s and Cauchy’s laws see Truesdell (1991).

The analysis of normal and shear stresses on families of surfaces at a point in a body is important in many engineering applications. Such an analysis can be carried out graphically using the technique of Mohr’s circles. For a discussion of this technique see Mase (1970) and Malvern (1969). In this respect we remark that, while we have defined the normal stress as the magnitude of the normal traction vector, many authors

define it as the normal component of the traction vector. These two definitions are equivalent up to sign.

Exercises

- 3.1 Consider a body with configuration $B = \{\mathbf{x} \in \mathbf{E}^3 \mid |x_i| < 1\}$ and mass density field $\rho = \exp(x_3)$. Find:
 (a) $\text{vol}[B]$, (b) $\mathbf{x}_{\text{cov}}[B]$, (c) $\text{mass}[B]$, (d) $\mathbf{x}_{\text{com}}[B]$.
- 3.2 Provide an example of a body B and corresponding mass density field ρ for which $\mathbf{x}_{\text{cov}}[B] \notin B$ and $\mathbf{x}_{\text{com}}[B] \notin B$.
- 3.3 Consider a body B with density field ρ and volume $\text{vol}[B] > 0$. The **inertia tensor** of B with respect to a point $\mathbf{y}$ is a second-order tensor $\mathbf{l}_{\mathbf{y}}$ defined by

$$\mathbf{l}_{\mathbf{y}} = \int_B \rho(\mathbf{x}) [|\mathbf{x} - \mathbf{y}|^2 \mathbf{I} - (\mathbf{x} - \mathbf{y}) \otimes (\mathbf{x} - \mathbf{y})] dV_{\mathbf{x}}.$$

- (a) Show that $\mathbf{l}_{\mathbf{y}}$ is symmetric, positive-definite for any point $\mathbf{y}$.
 (b) Let M be the total mass and $\bar{\mathbf{x}}$ the center of mass of B . Show that $\mathbf{l}_{\mathbf{y}}$ can be decomposed as

$$\mathbf{l}_{\mathbf{y}} = \mathbf{l}_{\bar{\mathbf{x}}} + M [|\bar{\mathbf{x}} - \mathbf{y}|^2 \mathbf{I} - (\bar{\mathbf{x}} - \mathbf{y}) \otimes (\bar{\mathbf{x}} - \mathbf{y})]$$

for any point $\mathbf{y}$.

- 3.4 Suppose a body with configuration $B = \{\mathbf{x} \in \mathbf{E}^3 \mid 0 < x_i < 1\}$ and constant mass density $\rho > 0$ is subject to a gravitational force field per unit mass $\mathbf{b} = -g\mathbf{e}_3$, where g is the gravitational acceleration constant. Find:
 (a) the resultant force $\mathbf{r}_b[B]$ (weight of B),
 (b) the resultant torque $\boldsymbol{\tau}_b[B]$ about the origin,
 (c) the resultant torque $\boldsymbol{\tau}_b[B]$ about the mass center $\mathbf{x}_{\text{com}}[B]$.
- 3.5 Consider an arbitrary body B subject to a body force.
- (a) Show that the resultant torque about the center of mass vanishes when the body is subject to a uniform (constant) body force field per unit mass. Does the resultant torque about any other reference point also vanish in this case?
- (b) Show that the resultant torque about the center of volume

vanishes when the body is subject to a uniform body force field per unit volume. Does the resultant torque about any other reference point also vanish in this case?

- 3.6 Suppose a body with configuration $B = \{\mathbf{x} \in \mathbb{E}^3 \mid 0 < x_i < 1\}$ is subject to a body force field per unit volume $\hat{\mathbf{b}} = \alpha x_3 \mathbf{e}_3$ and a traction field on its bounding surface ∂B given by

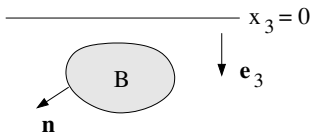
$$\mathbf{t} = \begin{cases} x_1 x_2 (1 - x_1)(1 - x_2) \mathbf{e}_3, & \text{on face } x_3 = 0, \\ \mathbf{0}, & \text{on all other faces.} \end{cases}$$

Find the value of α (constant) for which the resultant body and surface forces are balanced, that is, $\mathbf{r}_b[B] + \mathbf{r}_s[\partial B] = \mathbf{0}$.

- 3.7 Consider a solid body B immersed in a liquid of constant mass density ρ_* and subject to a uniform gravitational force field per unit mass. Suppose that the free surface of the liquid coincides with the plane $x_3 = 0$, and that the downward direction (into the liquid) coincides with $\mathbf{e}_3$. In this case, the liquid exerts a hydrostatic surface force field on the bounding surface of B

$$\mathbf{t} = -p\mathbf{n},$$

where $\mathbf{n}$ is the outward unit normal on the surface of B , $p = \rho_* g x_3$ is the hydrostatic pressure in the liquid, and g is the gravitational acceleration constant.



(a) Use the Divergence Theorem to show that the resultant hydrostatic surface force on B (the buoyant force) is given by $\mathbf{r}_s[\partial B] = -W\mathbf{e}_3$, where W is the weight of the liquid displaced by B .

(b) Show that the hydrostatic surface force has a zero resultant torque about the center of volume of B , that is, $\boldsymbol{\tau}_s[\partial B] = \mathbf{0}$.

Remark: The result in (a) is known as **Archimedes' Principle**. It states that the buoyant force on an object equals the weight of the displaced liquid. The result in (b) shows that the buoyant force acts at the center of volume of the object.

- 3.8 Prove that if $\mathbf{r}[\Omega]$ as defined in (3.5) vanishes, then $\boldsymbol{\tau}[\Omega]$ as defined in (3.6) is independent of $\mathbf{z}$.
- 3.9 Consider a body $B = \{\mathbf{x} \in \mathbf{E}^3 \mid 0 < x_i < 1\}$ with constant mass density $\rho > 0$ subject to a constant body force per unit mass $[\mathbf{b}] = (0, 0, -g)^T$. Suppose the Cauchy stress field in B is given by

$$[\mathbf{S}] = \begin{pmatrix} x_2 & x_3 & 0 \\ x_3 & x_1 & 0 \\ 0 & 0 & \rho g x_3 \end{pmatrix}.$$

- (a) Show that $\mathbf{S}$ and $\mathbf{b}$ satisfy the local equilibrium equations in Result 3.5.
- (b) Find the traction field on each of the six faces of the bounding surface ∂B .
- (c) Find by direct calculation the resultant surface force $\mathbf{r}_s[\partial B]$ and the resultant body force $\mathbf{r}_b[B]$ and verify that these forces are balanced, that is, $\mathbf{r}_s[\partial B] + \mathbf{r}_b[B] = \mathbf{0}$. Briefly explain how this result is consistent with part (a).
- 3.10 Consider a body B with uniform mass density field $\rho > 0$ (constant) under the influence of a body force per unit mass $\mathbf{b} = -4\mathbf{C}\mathbf{x}$, where $\mathbf{C}$ is a given second-order tensor (constant). Moreover, suppose the Cauchy stress field in B is of the form

$$\mathbf{S} = \rho (\mathbf{C}\mathbf{x}) \otimes \mathbf{x}.$$

- (a) Show that $\mathbf{S}$ and $\mathbf{b}$ satisfy the local equilibrium equation $\nabla \cdot \mathbf{S} + \rho \mathbf{b} = \mathbf{0}$ for balance of forces.
- (b) Find conditions on $\mathbf{C}$ for which the local equilibrium equation $\mathbf{S}^T = \mathbf{S}$ for balance of torques will be satisfied.
- 3.11 Consider a continuum body B with mass density ρ subject to a body force per unit mass $\mathbf{b}$ and a traction $\mathbf{h}$ on its bounding surface. Assume the Cauchy stress field $\mathbf{S}$ in B is related to a vector field $\mathbf{u} : B \rightarrow \mathcal{V}$ by the expression

$$\mathbf{S} = \mathbf{C}\mathbf{E} \quad \text{or} \quad S_{ij} = C_{ijkl}E_{kl},$$

where $\mathbf{C}$ is a constant fourth-order tensor and $\mathbf{E} : B \rightarrow \mathcal{V}^2$ is defined by

$$\mathbf{E} = \text{sym}(\nabla \mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T).$$

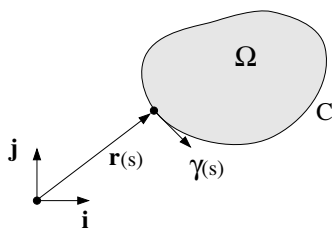
Here $\mathbf{u}$ is the **displacement field** of the body from an unstressed state and $\mathbf{E}$ is the **infinitesimal strain tensor**. Moreover, let W denote the **strain energy** in B , defined by

$$W = \frac{1}{2} \int_B \mathbf{E}(\mathbf{x}) : \mathbf{C} \mathbf{E}(\mathbf{x}) \, dV_{\mathbf{x}}.$$

Assuming B is in equilibrium show that

$$W = \frac{1}{2} \left(\int_B \rho(\mathbf{x}) \mathbf{b}(\mathbf{x}) \cdot \mathbf{u}(\mathbf{x}) \, dV_{\mathbf{x}} + \int_{\partial B} \mathbf{h}(\mathbf{x}) \cdot \mathbf{u}(\mathbf{x}) \, dA_{\mathbf{x}} \right).$$

- 3.12 Consider a straight bar B of uniform cross-section whose axis is parallel to the z -axis of an xyz -coordinate system. Let Ω denote a typical cross-section of B and assume the boundary $\partial\Omega$ is described by a smooth curve C in the xy -plane as illustrated in the figure below.



- (a) Let $\mathbf{r}(s) = x(s)\mathbf{e}_1 + y(s)\mathbf{e}_2$, $0 \leq s \leq L$, be an arclength parametrization of C , so that

$$\boldsymbol{\gamma} = \frac{d\mathbf{r}}{ds} = \frac{dx}{ds}\mathbf{e}_1 + \frac{dy}{ds}\mathbf{e}_2$$

is a unit tangent vector field on C in the direction of increasing arclength parameter s . Show that the vector field

$$\mathbf{n} = \frac{dy}{ds}\mathbf{i} - \frac{dx}{ds}\mathbf{j}$$

is a unit vector field on C , is everywhere orthogonal to $\boldsymbol{\gamma}$, and is oriented such that $\mathbf{n} \times \boldsymbol{\gamma} = \mathbf{e}_3$.

- (b) Suppose the ends of the bar B are twisted relative to each other by an amount so small that the configuration of B remains essentially unchanged; in particular, cross-sections do not warp and remain perpendicular to the z -axis. For such twisting, we

may assume that the Cauchy stress in B is of the form

$$[\mathbf{S}] = \begin{pmatrix} 0 & 0 & \tau_x \\ 0 & 0 & \tau_y \\ \tau_x & \tau_y & 0 \end{pmatrix},$$

where τ_x and τ_y are each functions of x and y only, and that these two functions are related to a single scalar function $\phi(x, y)$ by

$$\tau_x = \frac{\partial \phi}{\partial y}, \quad \tau_y = -\frac{\partial \phi}{\partial x}.$$

Determine the boundary condition imposed on the function ϕ by the requirement that $\mathbf{S}\mathbf{n} = \mathbf{0}$ on C . That is, what conditions must ϕ satisfy on C in order that the lateral surface of the bar be traction-free?

- 3.13 Consider a continuum body B subject to a body force per unit volume $\hat{\mathbf{b}}$ and a traction field $\mathbf{h}$ on its bounding surface. Suppose B is in equilibrium and let $\mathbf{S}$ denote its Cauchy stress field. Define the **average stress tensor** $\bar{\mathbf{S}}$ (a constant) in B by

$$\bar{\mathbf{S}} = \frac{1}{\text{vol}[B]} \int_B \mathbf{S} \, dV_{\mathbf{x}}.$$

- (a) Use the local equilibrium equations for B to show that

$$\bar{\mathbf{S}} = \frac{1}{\text{vol}[B]} \left[\int_B \mathbf{x} \otimes \hat{\mathbf{b}} \, dV_{\mathbf{x}} + \int_{\partial B} \mathbf{x} \otimes \mathbf{h} \, dA_{\mathbf{x}} \right].$$

- (b) Suppose that $\hat{\mathbf{b}} = \mathbf{0}$ and that the applied traction $\mathbf{h}$ is a uniform pressure, so that

$$\mathbf{h} = -p\mathbf{n},$$

where p is a constant and $\mathbf{n}$ is the outward unit normal field on ∂B . Use the result in (a) to show that the average stress tensor is spherical, namely

$$\bar{\mathbf{S}} = -p\mathbf{I}.$$

- (c) Under the same conditions as in (b) show that the uniform, spherical stress field $\mathbf{S} = -p\mathbf{I}$ satisfies the equations of equilibrium in B and the boundary condition $\mathbf{S}\mathbf{n} = \mathbf{h}$ on ∂B . Thus in this case we have $\mathbf{S}(\mathbf{x}) = \bar{\mathbf{S}}$ for all $\mathbf{x} \in B$.

Remark: The result in (a) is known as **Signorini's Theorem**. It states that the average value of the Cauchy stress tensor in a body in equilibrium is completely determined by the external surface traction $\mathbf{h}$ and the body force $\widehat{\mathbf{b}}$.

- 3.14 Show that uniaxial and pure shear stress states are examples of plane stress states. In particular, for each of these states find a basis in which the matrix representation $[\mathbf{S}]$ has the required form.
- 3.15 Suppose the Cauchy stress tensor at a point $\mathbf{x}$ in a body has the form

$$[\mathbf{S}] = \begin{pmatrix} 5 & 3 & -3 \\ 3 & 0 & 2 \\ -3 & 2 & 0 \end{pmatrix},$$

and consider a surface Γ with normal $[\mathbf{n}] = (0, 1/\sqrt{2}, 1/\sqrt{2})^T$ and a surface Γ' with normal $[\mathbf{n}'] = (1, 0, 0)^T$ at $\mathbf{x}$.

(a) Find the normal and shear tractions $[\mathbf{t}_n]$ and $[\mathbf{t}_s]$ on each surface at $\mathbf{x}$. In particular, show that Γ experiences no shear traction at $\mathbf{x}$, whereas Γ' does.

(b) Find the principal stresses and stress directions at $\mathbf{x}$ and verify that $[\mathbf{n}]$ is a principal direction.

- 3.16 Suppose the Cauchy stress field in a body $B = \{\mathbf{x} \in \mathbf{E}^3 \mid |x_i| < 1\}$ is uniaxial of the form

$$[\mathbf{S}] = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

where $\sigma \neq 0$ is constant. In this case notice that the traction field $\mathbf{t}$ on any plane through B will be constant because $\mathbf{S}$ and $\mathbf{n}$ are constant.

(a) Consider the family of planes Γ_θ through the origin which contain the x_1 -axis and have unit normal $[\mathbf{n}] = (0, \cos \theta, \sin \theta)^T$, $\theta \in [0, \pi/2]$. Find the normal and shear stresses σ_n and σ_s on these planes as a function of θ .

(b) Show that the maximum normal stress is $\sigma_n = |\sigma|$ and that this value occurs on the plane with $\theta = 0$. Similarly, show that the maximum shear stress is $\sigma_s = \frac{1}{2}|\sigma|$ and that this value occurs on the plane with $\theta = \pi/4$.

Remark: The result in (b) illustrates the principle that planes of maximum shear stress occur at 45-degree angles to planes of maximum normal stress.

3.17 Prove Result 3.6 (1).

3.18 Prove Result 3.6 (2).

3.19 Let p be the pressure at a point $\mathbf{x}$ in a continuum body and let $\mathcal{N}$ be the set of all unit vectors $\mathbf{n}$. By identifying $\mathcal{N}$ with the standard unit sphere show

$$p = -\frac{\int_{\mathcal{N}} \mathbf{n} \cdot \mathbf{t} dA_{\mathbf{n}}}{\int_{\mathcal{N}} dA_{\mathbf{n}}},$$

where $\mathbf{t}$ is the traction vector on a surface with normal $\mathbf{n}$ at $\mathbf{x}$.

3.20 Two different measures of stress intensity in a body are provided by the functions $f(\mathbf{S}) = |p|$ and $g(\mathbf{S}) = \sqrt{J_2}$, where p is the pressure and J_2 is the second deviatoric invariant associated with the Cauchy stress field $\mathbf{S}$. Find f and g for the following stress states $[\mathbf{S}]$:

$$(a) \begin{pmatrix} \sigma & 0 & 0 \\ 0 & \sigma & 0 \\ 0 & 0 & \sigma \end{pmatrix}, \quad (b) \begin{pmatrix} 0 & \tau & 0 \\ \tau & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad (c) \begin{pmatrix} \sigma & 0 & 0 \\ 0 & -\sigma & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Remark: The function $g(\mathbf{S}) = \sqrt{J_2}$ is typically called the **Mises yield function** and is employed in the modeling of rigid-plastic materials. Such materials are considered to be rigid, and deform only when $g(\mathbf{S})$ exceeds a threshold value.

Answers to Selected Exercises

3.1 Let $e = \exp(1)$ be the base of the natural logarithm. Then:

$$(a) \text{vol}[B] = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 dx_1 dx_2 dx_3 = 8.$$

$$(b) \mathbf{x}_{\text{cov}}[B] = \frac{1}{\text{vol}[B]} \left(\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 x_i dx_1 dx_2 dx_3 \right) \mathbf{e}_i = \mathbf{0}.$$

$$(c) \text{mass}[B] = \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 e^{x_3} dx_1 dx_2 dx_3 = 4e - 4e^{-1}.$$

$$(d) \mathbf{x}_{\text{com}}[B] = \frac{1}{\text{mass}[B]} \left(\int_{-1}^1 \int_{-1}^1 \int_{-1}^1 e^{x_3} x_i dx_1 dx_2 dx_3 \right) \mathbf{e}_i = \left(\frac{2}{e^2 - 1} \right) \mathbf{e}_3.$$